

The vertical migration step is a unit translation, and the orientation gap is irreducible

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Abstract

In `proofs/2026-10-05-c3-migration-length-grading.tex` the migration length $\ell(M)$ of Purbhoo's mosaic migration (0705.1184) was shown to admit an exact potential $\varphi(a, b, c, e) = b + e$ on the $\mathbb{Z}^4$ lift, with $\Delta\varphi = 1$ at every step — except that the strictly vertical 180° step exists in two directions, and 91 of 707 computed steps were strictly vertical with all 91 taking $\Delta\varphi = +1$ and no proof excluding -1 . This note sharpens that gap. We show the two vertical steps are one geometric picture read in two directions (Proposition 1), deduce that *no* choice of potential can assign $+1$ to both (Corollary 2), so the gap can only be closed by a reachability argument; we show hexagon minimality cannot discriminate (Proposition 3); and we widen and, more importantly, *explain* the empirical null, from 707 to 3111 steps. The gap is not closed. Two hypotheses on which Theorem 1 of the earlier paper rests are stated explicitly, one of them newly identified.

1 Setting

Vertices of a mosaic are integer 4-tuples (a, b, c, e) standing for $a u_0 + b u_{60} + c u_{30} + e u_{90}$, where u_t is the unit vector at angle t ; the map $\mathbb{Z}^4 \rightarrow \mathbb{R}^2$ is injective. Set $\varphi(a, b, c, e) = b + e$, extended to a tile as the mean of its four vertices. Migration travels in the direction $u = u_{150}$ (the north direction N_A of nest A), and we write $\sigma = \langle \cdot, u_{150} \rangle$ for the travel coordinate. Purbhoo's step rotates the smallest hexagon containing the travelling rhombus and lying weakly right of its leftmost point, by 180° (centrally symmetric hexagon) or $\pm 120^\circ$ (three-fold symmetric hexagon). A step is *strictly vertical* when σ is unchanged.

2 The vertical step is a unit translation

Proposition 1. *Up to translation there are exactly two strictly vertical migration steps, and they share the same hexagon region*

$$H = R \oplus [0, 1] u_{60},$$

the Minkowski sum of the travelling rhombus R with a unit segment perpendicular to the travel direction. H is the zonogon on $\{u_0, u_{60}, u_{150}\}$, has four tiles, and has exactly two tilings, interchanged by its central symmetry. The 180° rotation translates the rhombus by $+u_{60}$ in one tiling and by $-u_{60}$ in the other, giving $\Delta\varphi = \varphi(u_{60}) = +1$ and -1 respectively.

Proof. Exhaustive enumeration of every admissible hexagon up to translation (90 regions), every tiling, every rhombus and every admissible rotation, filtered by Purbhoo’s constraints (the rhombus attains the hexagon’s σ -minimum; the rotated rhombus keeps one of the two orientations of Proposition 3.1), returns exactly two steps with σ unchanged. Both have the hexagon with vertex cycle $(0, 0, -1, 1), (0, 1, -1, 1), (1, 1, -1, 1), (1, 1, 0, 0), (1, 0, 0, 0), (0, 0, 0, 0)$, i.e. the zonogon on u_0, u_{60}, u_{150} : of its three pairs of edge directions, $\{u_0, u_{60}\}$ spans a 60° rhombus (two triangles), $\{u_0, u_{150}\}$ the $30/150$ rhombus tile and $\{u_{60}, u_{150}\}$ a square, so $|H| = 4$ tiles. In one tiling the rhombus is $R = \{(0, 0, -1, 1), (0, 0, 0, 0), (1, 0, 0, 0), (1, 0, -1, 1)\}$ and in the other it is $R + (0, 1, 0, 0) = R + u_{60}$, and these are the two tilings of a three-direction zonogon. Since $u_{60} \perp u_{150}$, both have the same σ -extent, which is why the step is vertical; and φ is $\mathbb{Z}$ -linear with $\varphi(u_{60}) = 1$, so $\Delta\varphi = \pm 1$ according to the direction. $\square$

Corollary 2. *There is no function ψ on rhombus positions with $\Delta\psi = +1$ on both vertical steps. Hence the gap cannot be closed by replacing φ with a better potential; it can be closed only by showing that one of the two directions is never realised.*

Proof. By Proposition 1 the two steps are mutually inverse: the same region, the same rotation, the two tilings. So for any ψ , $\Delta\psi(T_+ \rightarrow T_-) = -\Delta\psi(T_- \rightarrow T_+)$, and the two cannot both equal $+1$. $\square$

Proposition 3. *Minimality of the hexagon does not discriminate between the two directions.*

Proof. Enumerating every admissible local move for the canonical nest- A rhombus gives 15 moves, and *every one of them* uses a four-tile hexagon: one vertical with $\Delta\varphi = -1$, one vertical with $+1$, one forward two-fold with $+1$, six backward three-fold with -1 (excluded by forward motion) and six forward three-fold with $+1$. Since all candidate hexagons have the same size, “the smallest hexagon containing the rhombus” cannot prefer one vertical direction over the other. Note also that $H = R \oplus [0, 1]u_{60}$ has the same σ -extent as R , so the clause “lying weakly right of the rhombus’s leftmost point” is satisfied by *both* strips and is likewise silent. $\square$

So the asymmetry, if any, is global. Proposition 1 does localise it to a concrete statement: the 180° rotation maps the strip’s rear end to its front end, so the two strips differ in where the square sits relative to travel.

Hypothesis 4 (G). In a mosaic the travelling rhombus never has a free unit strip on its $-u_{60}$ side; equivalently, it never has the configuration (*square ahead, two triangles*) occupying $R \ominus [0, 1]u_{60}$.

3 The null, widened and explained

The earlier figure was 91 of 707 steps strictly vertical, all $+1$. That count alone was not evidence. The step engine breaks a tie among equal-progress candidates by maximising progress, and progress is *constant* on vertical candidates, so the tie-break returns whichever candidate the enumeration happened to list first. Had the -1 slide ever been in the candidate pool, an all- $+1$ tally would have been an artifact of enumeration order and not a fact about mosaics. So availability was measured, not selection, over $(n, d) = (4, 2), (5, 2), (5, 3), (6, 3)$: 362 mosaics, 3111 steps.

steps logged	3111
chosen step strictly vertical	434
of those, $\Delta\varphi = +1$	434
of those, $\Delta\varphi = -1$	0
steps where a vertical $\Delta\varphi = -1$ candidate was available	0
steps where a vertical $\Delta\varphi = +1$ candidate was available	434
steps where both were available	0

The silence is *absence*, not deselection: the tie-break is never consulted on a vertical step, so the null is a statement about mosaic geometry rather than about the implementation. Hypothesis 4 is therefore an empirical null at 3111 steps and 362 mosaics, and remains unproved.

4 A second hypothesis, newly identified

The exhaustive classification underpinning the potential filters on *exactly one rhombus in the hexagon*; the engine imposes no such restriction. Measuring what the engine actually does:

chosen steps	3111
with $\Delta\varphi = 1$ (bare height, no type correction)	3111
whose chosen hexagon contains exactly one rhombus	3111
whose chosen hexagon has exactly four tiles	3111

So the classification does cover every chosen step in this range — but “exactly one rhombus” is a property of the *chosen* hexagon, not a consequence of the stated rule, and the potential identity needs it:

Hypothesis 5 (G2). The minimal admissible hexagon of the travelling rhombus contains no other rhombus.

This is load-bearing, because the candidate pool *did* contain moves violating the potential identity: 173 candidates with bare $\Delta\varphi = +\frac{1}{2}$ and 173 with $-\frac{1}{2}$. They were always rejected, and the reason is now established rather than assumed: all 346 rejected candidates had rhombus orientation (120, 90), which Purbhoo’s Proposition 3.1 forbids, and all 346 had the horizontal/vertical flag false, while all 346 chosen moves had it true. Both controls returned zero (no rejected candidate had an allowed orientation; no chosen candidate had a forbidden one). Hence:

Proposition 6. *On these 3111 steps, Purbhoo’s horizontal/vertical tie-break selects exactly the orientations permitted by his Proposition 3.1. The tie-break is not an independent heuristic; it implements the orientation restriction.*

A byproduct: no rhombus-type correction is needed in the potential. The bare height φ has $\Delta\varphi = 1$ on every chosen step, all 3111.

Finally, the move repertoire of a journey: 1130 two-fold steps $(0, 150) \rightarrow (0, 150)$, 696 two-fold $(60, 30) \rightarrow (60, 30)$, 1112 three-fold $(0, 150) \rightarrow (60, 30)$ and 173 three-fold $(60, 30) \rightarrow (0, 150)$. A journey slides within an orientation by 180° rotations and alternates orientation by three-fold rotations, starting in nest A ’s orientation (spanned by $E_A = u_0$, $N_A = u_{150}$) and ending in nest B ’s (spanned by $E_B = u_{240}$, $N_B = u_{30}$).

Gaps

1. Hypotheses 4 and 5 are unproved. Theorem 1 of `2026-10-05-c3-migration-length-grading.tex` must state both as hypotheses; it previously stated neither explicitly, (G2) being hidden in a filter of the classification script.
2. Corollary 2 rules out the entire family of repairs by a better potential, so no further search in that direction is warranted.
3. The reach is unchanged on one axis: no fibre with $c_{\mu\nu}^{\lambda} \geq 3$ was run (the tiler is exponential); $c \geq 2$ was exercised on 21 fibres. The 3111-step sweep is not exhaustive in the multiplicity direction.

Apparatus

`work-in-progress/code/mosaic-migration-1005c3/` (unchanged), driven by the measurement scripts of this session.